

OOPS ASSIGNMENT

Question1.

Code :

```
#include <iostream>
using namespace std;
class Sample {
int x;
public:
void set (int x) {
this->x = x;
}
void show () {
cout<< "Value of x = " << x <<
endl;
}
};
int main() {
Sample obj;
Sample *ptr;
ptr = &obj;
obj.set(40);
obj.show();
ptr->set(50);
ptr->show();
return 0;
}
```

Output:

```
Value of x = 40  
Value of x = 50  
  
...Program finished with exit code 0
```

Question2.

Code:

```
#include <iostream>
using namespace std;
class B;
class A {
int x;
public:
A(int a) { x = a; }
friend void swap(A&, B&);
void show() { cout << "A = " << x <<
endl; }
};
class B {
int y;
public:
B(int b) { y = b; }
friend void swap(A&, B&);

void show() { cout << "B = " << y <<
endl; }
};
void swap(A &a, B &b) {
int temp = a.x;
a.x = b.y;
b.y = temp;
}
int main() {
A a(23);
B b(42);
swap(a, b);
a.show();
b.show();
return 0;
}
```

Output:

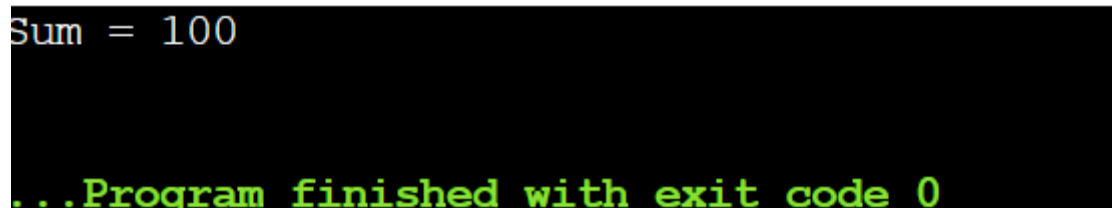
```
A = 42
B = 23
```

```
...Program finished with exit code 0
```

Question3.

Code:

```
#include <iostream>
using namespace std;
class B;
class A {
int x;
public:
A(int a) { x = a; }
friend int add(A, B);
};
class B {
int y;
public:
B(int b) { y = b; }
friend int add(A, B);
};
int add(A a, B b) {
return a.x + b.y;
}
int main() {
A a(78);
B b(22);
cout << "Sum = " << add(a, b) <<
endl;
return 0;
}Output:
```



Sum = 100

...Program finished with exit code 0

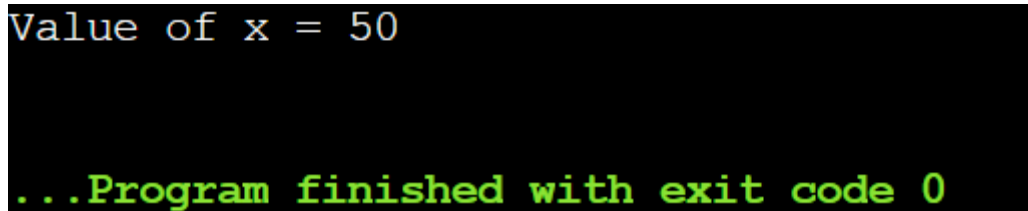
Question4.

Code:

```
#include <iostream>
using namespace std;
class A {
int x;
public:
A() { x = 50; }
friend class B;
};
class B {
public:
void show(A a) {
cout << "Value of x = " << a.x <<
endl;
}
```

```
};  
int main() {  
    A a;  
    B b;  
    b.show(a);  
    return 0;  
}
```

Output:



```
Value of x = 50  
  
...Program finished with exit code 0
```

Question5.

Code:

```
#include <iostream>  
  
using namespace std;  
  
class Rectangle {  
    int length, breadth;  
  
public:  
    void set() {  
        cin >> length >> breadth;  
    }  
  
    void area() {  
        cout << "Area = " << length *  
        breadth << endl;  
    }  
};  
  
int main() {  
    Rectangle r[3];  
  
    for(int i = 0; i < 3; i++) {  
        cout << "Enter length and breadth:  
        ";  
        r[i].set();  
    }  
  
    for(int i = 0; i < 3; i++) {  
        r[i].area();  
    }  
}
```

```
}  
return 0;  
}
```

Output:

```
Enter length and breadth:12 30  
Enter length and breadth:23 43  
Enter length and breadth:23 12  
Area = 360  
Area = 989  
Area = 276  
  
...Program finished with exit code 0
```

Question 6:

Code:

```
#include <iostream>  
using namespace std;  
inline int cube(int x) {  
    return x * x * x;  
}  
int main() {  
    int n;  
    cout << "Enter number: ";  
    cin >> n;  
    cout << "Cube = " << cube(n) <<  
    endl;  
    return 0;  
}
```

Output:

```
Enter number: 23  
Cube = 12167  
  
...Program finished with exit code 0
```

Question7.

(a)

Code:

```
#include <iostream>
using namespace std;
class Number {
public:
int x;
};
Number modify(Number n) {
n.x += 10;
return n;
}
int main() {
Number n1, n2;
n1.x = 5;
n2 = modify(n1);
cout << "Original: " << n1.x <<
endl;
cout << "Modified: " << n2.x <<
endl;
return 0;
}
```

Output:

```
Original: 5
Modified: 15

...Program finished with exit code 0
```

(b)

Code:

```
#include <iostream>
using namespace std;
class Number {
public:
int x;
};
Number modify(Number *n) {
n->x += 10;
return *n;
}
int main() {
Number n1, n2;
n1.x = 5;
n2 = modify(&n1);
cout << "Modified: " << n2.x << endl;
return 0;
}
```

Output:

```
Modified: 15

...Program finished with exit code 0
```